

Shantanu Gorai

+91 70619 36584 | shantanugorai013@gmail.com | linkedin.com/in/shantanu-gorai | github.com/ShantanuGorai

SUMMARY

Computer Science professional with hands-on proficiency in Python, RESTful API architecture, and database diagnostics. Experienced in application troubleshooting, root-cause analysis (RCA), and debugging multi-tier distributed systems. Adept at acting as an investigative software specialist to diagnose production issues, optimize system uptime, and support data-driven and AI-integrated business workflows.

TECHNICAL SKILLS

Programming & Scripting:	Python (Advanced), JavaScript (ES6+), TypeScript, SQL, Bash/Shell Scripting
Application Support & Ops:	Root Cause Analysis (RCA), Log Analysis, Incident Management, API Debugging, Monitoring
Backend & APIs:	Flask, Node.js, Express.js, RESTful APIs, WebSockets, JWT, Microservices
AI & Data Solutions:	Generative AI Concepts, LLM API Integration, ML Model Deployment, Data Pipelines
Databases & Platforms:	PostgreSQL, MySQL, MongoDB, Linux/Unix, Docker, Git, Postman, Vercel
Frontend Web Tech:	React.js, Next.js, HTML5, CSS3, Tailwind CSS

EXPERIENCE

Vocational Trainee — Machine Learning & Digital Systems	Apr 2026 – May 2026
<i>Tata Steel Ltd., Learning & Development Department</i>	<i>Jamshedpur, Jharkhand</i>
- Diagnosed data pipeline bottlenecks and debugged computer vision inference workflows in Python to ensure reliable image recognition operations. - Conducted root-cause analysis on system exceptions, performed regression testing, and documented support runbooks for plant stakeholders. - Monitored end-to-end service health across input ingestion and model prediction services to reduce downtime.	

FEATURED PROJECTS

Real-Time Network Monitoring & Threat Triage System <i>Python, Flask, WebSockets, React.js, Linux</i>	2025
- Engineered a real-time diagnostics dashboard tracking network traffic anomalies, packet drops, and threat events using Python packet sniffers and WebSockets. - Implemented automated error logging and structured alerting mechanisms to facilitate rapid root-cause isolation and incident response. - Troubleshooted performance degradation under high event throughput by optimizing multi-threaded data processing handlers.	
VocalLock — Secure User Authentication & Service Gateway <i>Python, Flask, MySQL, REST API, React.js</i>	2025
- Designed and debugged a secure multi-tier authentication platform handling session validations, role-based access, and token lifecycle management. - Analyzed API error logs via Postman and Flask debug loggers to resolve database deadlocks and invalid payload edge cases. Runner-Up, CODE CARNAGE National Hackathon (Apr 2025) — Built, debugged, and deployed a production-grade application under a strict 24-hour sprint.	
AI Smart Attendance & Academic Analytics Portal <i>Python, Flask, PostgreSQL, Generative AI APIs</i>	2024
- Developed an automated reporting service integrating AI summarization pipelines for automated anomaly detection in student attendance records. - Resolved complex PostgreSQL schema bottlenecks and query execution timeouts, improving data retrieval speeds for multi-user concurrent access. - Conducted continuous end-to-end integration testing between frontend states and backend API error handlers.	
Multi-Tenant Property Listing Platform <i>Node.js, Express.js, MongoDB, React.js</i>	2025
- Maintained backend CRUD operations, indexing strategies, and API routes to resolve pagination failures and high-latency data queries. - Implemented structured HTTP status code handling and client-side error boundaries to improve application resilience.	

EDUCATION

Bachelor of Engineering – Computer Science	2023 – 2027
<i>ACS College of Engineering, Bengaluru (VTU Affiliated)</i>	<i>Bengaluru, Karnataka</i>

ACHIEVEMENTS & HACKATHONS

- Runner-Up, CODE CARNAGE – National Hackathon (Apr 2025) | Semi-Finalist, CODE BREAKER Challenge (May 2025) | Top-5 Finalist, AI-VERSE 2.0 (Oct 2025)**